2026-04-03

Technology

- Language: Go (module `chat_sec`, see `go.mod`)
- UI: Gio (gioui.org) for the desktop client
- Networking: TCP sockets
- Crypto: Custom RSA implementation in `server/enc` (keygen, encrypt/decrypt, base64 marshaling)

Project Structure

- `main.go`: Gio desktop chat client (GUI)
- `server/main.go`: TCP chat server (broadcast + RSA decrypt)
- `server/enc/ma_rsa.go`: RSA implementation used by server and clients
- `assets/`: UI screenshots
- `main`, `client/client`, `server/server`, `chat_sec`: built binaries
- `go.mod`, `go.sum`: Go module metadata

How Does the Server Works

- On startup, the server generates a 2048-bit RSA key pair and starts listening on TCP port `9001`.
- When a client connects, the server sends `PUBKEY:<base64(N)|base64(E)>` as the first line.
- Client sends `USERNAME:<name>`; server stores the connection and broadcasts a system join message.
- Each client message is sent as `MSG:<base64(ciphertext)>`.
- Server base64-decodes the ciphertext, decrypts with the RSA private key, and broadcasts the plaintext to all clients as `<username>: <message>`.
- Disconnects remove clients and broadcast a system leave message.

Client See

- GUI client (`main.go`) opens a window with a username entry screen, then a chat view.
- Chat view shows a header (username + status), a scrollable message list, and an input bar with a Send button.
- Outgoing messages are encrypted with the server public key and sent as `MSG:` payloads.
- CLI client (`client/main.go`) prompts for username, prints incoming messages, and sends encrypted text on each line.

```
go run server/main.go
```

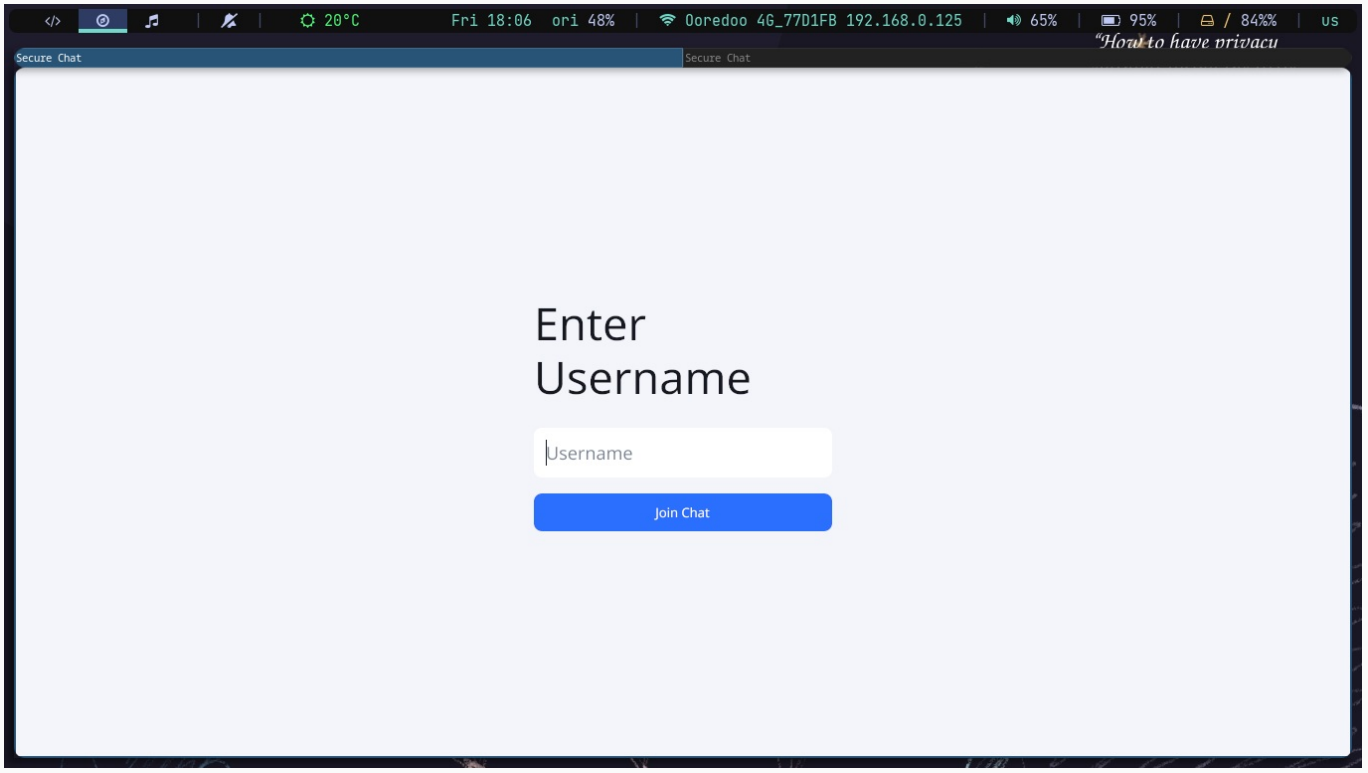
In another terminal:

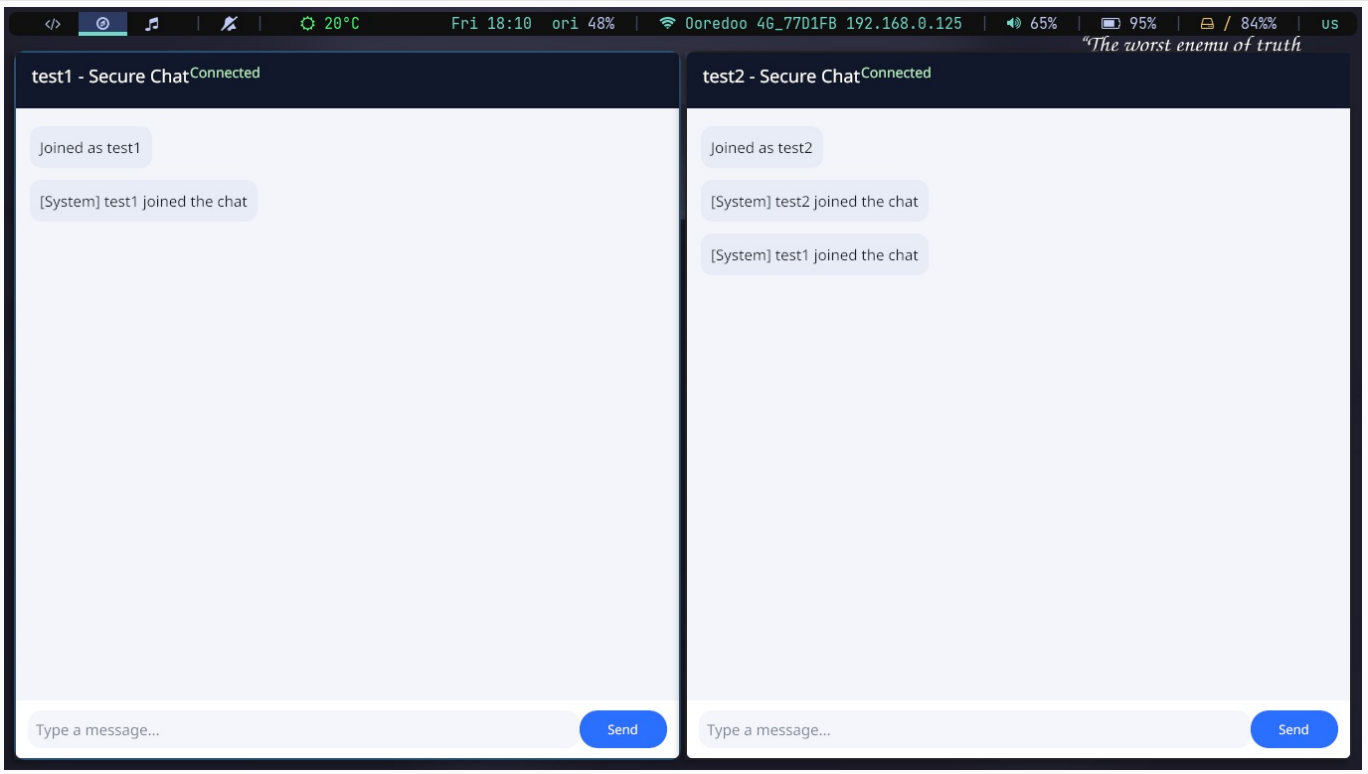
```
go run main.go     # GUI client
```

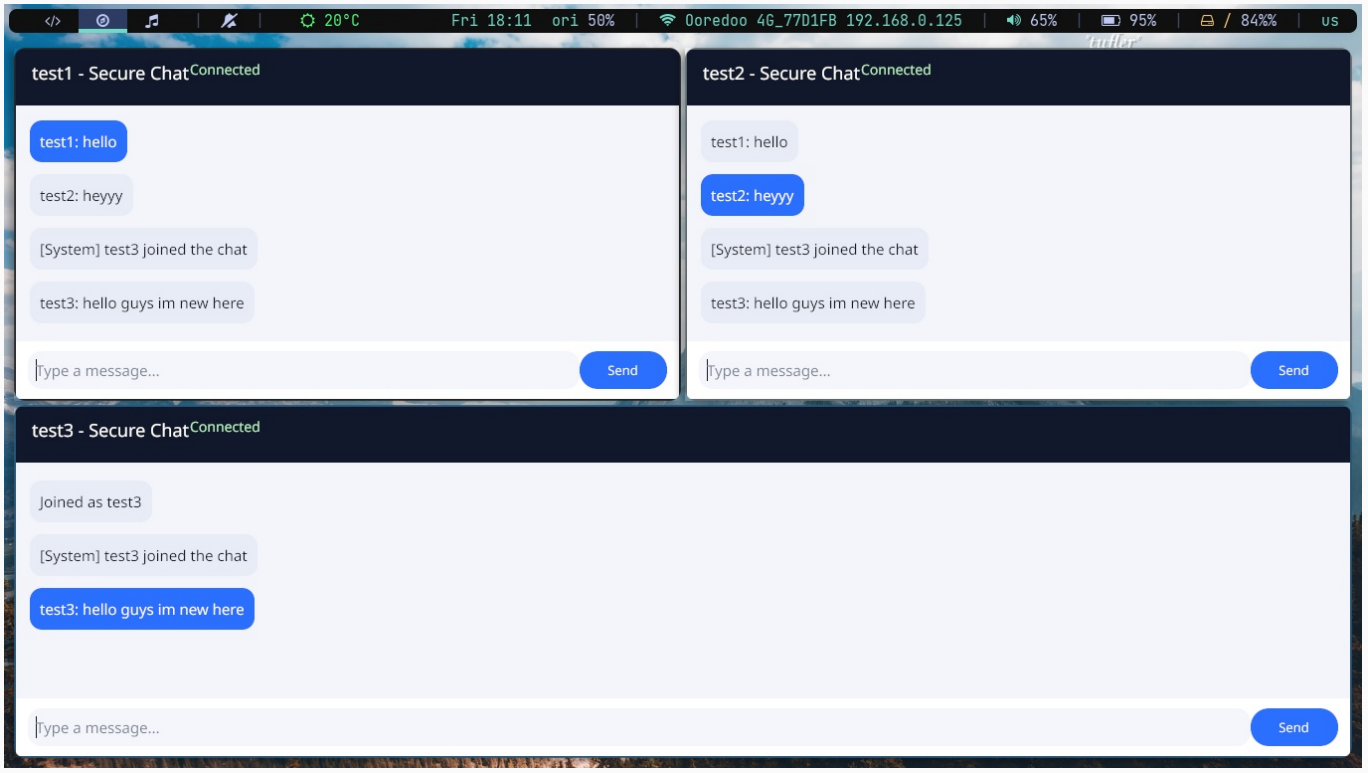
Notes: - Default host/port: `localhost:9001` in both clients and server. - Message size is limited by RSA key size (2048-bit key → max plaintext ~245 bytes in this implementation). - The RSA implementation is raw (no padding/auth). It is fine for a demo but not secure for production.

The View

Screenshots from `assets/`:


20°C
Fri 18:06 ori 48%
Ooredoo 4G_77D1FB 192.168.0.125
65%
95%
/ 84%%
us
"How to have privacy
Secure Chat
Secure Chat
Enter Username
Username
Join Chat


20°C
Fri 18:10 ori 48%
Ooredoo 4G_77D1FB 192.168.0.125
65%
95%
/ 84%%
us
"The worst enemy of truth
test1 - Secure Chat Connected
Joined as test1
[System] test1 joined the chat
Type a message...
Send
test2 - Secure Chat Connected
Joined as test2
[System] test2 joined the chat
[System] test1 joined the chat
Type a message...
Send


20°C
Fri 18:11 ori 50%
Ooredoo 4G_77D1FB 192.168.0.125
65%
95%
/ 84%%
us
test1 - Secure Chat Connected
test1: hello
test2: heyyy
[System] test3 joined the chat
test3: hello guys im new here
Type a message...
Send
test2 - Secure Chat Connected
test1: hello
test2: heyyy
[System] test3 joined the chat
test3: hello guys im new here
Type a message...
Send
test3 - Secure Chat Connected
Joined as test3
[System] test3 joined the chat
test3: hello guys im new here
Type a message...
Send